

Minhyeok (Shawn) Son

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EDUCATION

University of Michigan, Ann Arbor

Ann Arbor, MI

Master of Science in Data Science

Expected May 2027

- **Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Statistical Inference, Distributed Systems, Data Structures

EXPERIENCE

eBay

May 2026 – Aug 2026

Machine Learning Engineer Intern

San Jose, CA

- Enabled analysts to track **25** Compliance signal types from a single query via an LLM workflow in Python.
- Automated root-cause analysis by linking **3** classes of code events (configs, refreshes, deploys) to anomalies.
- Cut engineer escalations by **80%** (**10 hrs/week**) by enabling self-service diagnostics for analysts.
- Selected the production setup by benchmarking **15** model/prompt configs on accuracy, latency, and cost.
- Safeguarded production models by writing **150+** automated tests, catching **95%** of regressions pre-deploy.

American Airlines

Sep. 2025 – Nov. 2025

Applied Data Scientist

Ann Arbor, MI

- Unified **1M+** flight legs into a PostgreSQL feature store by joining operations, weather, and congestion data.
- Captured key delay drivers by engineering **50** weather, schedule, and airport-congestion features.
- Beat baselines by **8 AUC pts** on held-out flights by training and tuning gradient-boosted tree models.
- Served real-time delay forecasts via a Flask API at <100ms latency, handling **50K** daily calls.

Industrial AI Lab, Seoul National Univ. of Science & Technology

Sep. 2022 – Dec. 2024

Research Assistant

Seoul, South Korea

- Detected rare faults in a **2.9M**-row, 1-second heat-treatment stream by applying XGBoost, thresholds, and SHAP.
- Cut inspection cost on **~263K** semiconductor units (<**2%** fault) by co-developing a distillation framework.
- Hit near-advanced accuracy from low-cost features; uncertainty sampling beat random by **3 AUROC pts**.
- Published in *Scientific Reports* by contributing methodology and analysis across **30+** repeated trials.

PROJECTS

German E-commerce NER | 3rd of 80 · Solo

eval.ai/web/challenges

- Extracted entities from **10K** German e-commerce titles via an XLM-R / mDeBERTa + word-level CRF pipeline.
- Placed **3rd of 80** graduate teams competing solo at weighted-F3 **0.9466**, within **0.002** of 1st place.
- Stabilized predictions on noisy text by adding **100+** rule/lexicon priors and a verifier-gating ensemble.
- Surpassed **eBay's own baseline** by aligning validation to official scoring via a custom weighted-F3 metric.

Agentic RAG Security Assistant | Python, Go, PyTorch, LangChain

github.com/Shawn/Agentic-RAG

- Built a RAG-based ranking and retrieval system over **500K** security logs, lifting top-5 recall **22%**.
- Fine-tuned an open-source LLM on **50K** labeled alerts, lifting classification F1 by **12** points.
- Built a reasoning LLM agent that autonomously triaged **200** security alerts daily via tool-calling.
- Built a Go microservice streaming **3M** events/day from S3 to the inference pipeline at **99.9%** uptime.

PUBLICATION

Heo, J., Son, M., & Shim, J. "Active Inspection with Knowledge Distillation for Cost-Effective Fault Prediction in Manufacturing Process." *Scientific Reports*, 16, 8613 (2026).

TECHNICAL SKILLS

Languages: Python, Go, SQL (PostgreSQL), Java

ML / DL: PyTorch, TensorFlow, LLMs & RAG, Agentic AI, scikit-learn, Transformers / NLP, XGBoost, time series, knowledge distillation

Data & Tools: S3, Git, Docker, pandas, NumPy, SHAP, Artifactory, Jira, Streamlit

Work Authorization: F-1 visa — CPT-authorized for 2026 internships; STEM-OPT eligible for full-time (3 years)